

L4: The Maximum Degree in a Power-law Network

Let us now derive an expression for the maximum degree we can expect to see in a power-law network with n nodes, and to compare that with the corresponding maximum degree we can expect to get from a network in which the degree distribution decays exponentially fast with the degree k . Recall that the Poisson distribution of $G(n, p)$ networks decays even faster than exponential distribution.

To make the calculations easier, let us work with the (continuous) exponential distribution $p_k = c e^{-\lambda k}$, where k is the degree, $\frac{1}{\lambda}$ is the average degree, and c is a normalization constant. In a network in which the minimum degree is $k_{min} \geq 0$ (with $\frac{1}{\lambda} > k_{min}$), the parameter c should be equal to $\lambda e^{\lambda k_{min}}$, and so the probability that a node has degree k is

$$p_k = \lambda e^{\lambda k_{min}} e^{-\lambda k}$$

Suppose that the node with maximum degree k_{max} is unique – so the probability of having a degree k_{max} is $\frac{1}{n}$.

Thus, relying on the continuous k approximation again, we can write that:

$$\int_{k_{max}}^{\infty} p_k dk = \frac{1}{n}$$

Substituting the previous expression for p_k and calculating this integral, we can easily get that the maximum degree for the exponential degree distribution is:

$$k_{max} = k_{min} + \frac{\ln n}{\lambda}$$

This means that the **maximum degree increases very slowly (logarithmically) with the network size n , when the degree distribution decays exponentially fast with k .**

Let us now repeat these derivations but for a power-law network with the same minimum degree k_{min} and an exponent α .

$$p_k = \frac{\alpha-1}{k_{min}} \left(\frac{k}{k_{min}}\right)^{-\alpha}$$

Suppose that the node with maximum degree k_{max} is unique – so the probability of having a degree k_{max} is $\frac{1}{n}$.

Thus, relying on the continuous k approximation, we can again write that:

$$\int_{k_{max}}^{\infty} p_k dk = \frac{1}{n}$$

Substituting the previous expression for p_k and calculating this integral, we can easily get that the maximum degree for a power-law network is:

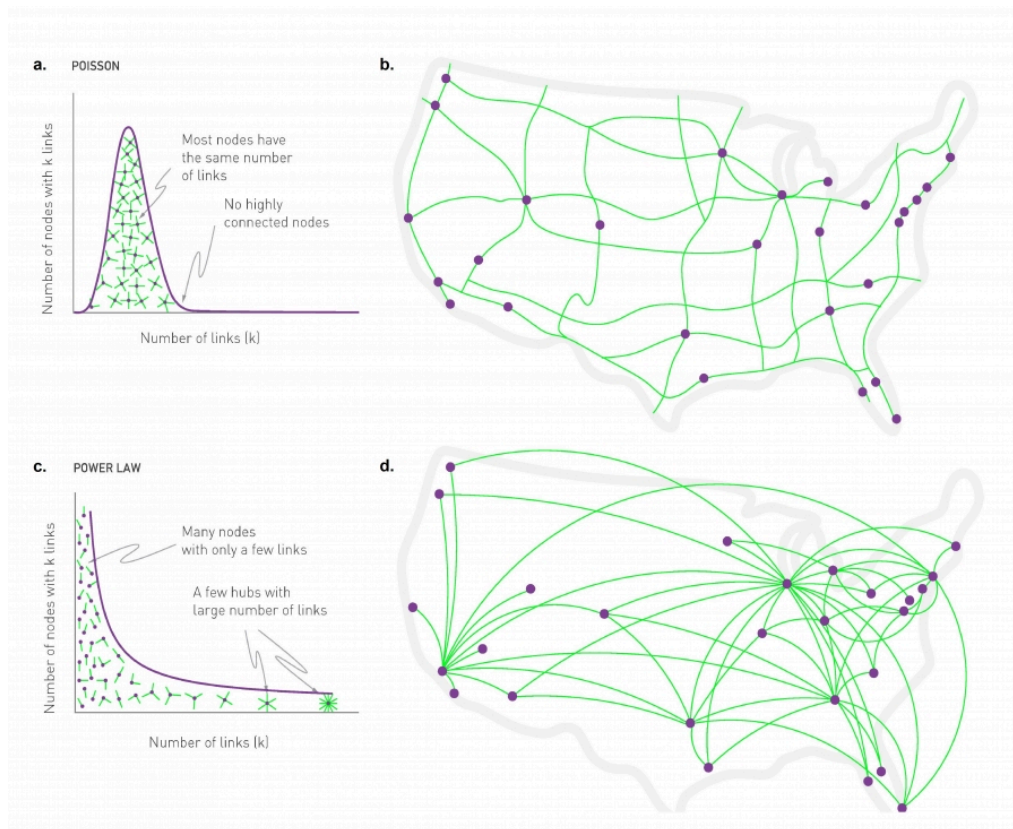
$$k_{max} = k_{min} n^{1/(\alpha-1)}$$

This means that the **maximum degree in a power-law network increases as a power-law of the network size n . If $\alpha = 3$ the maximum degree increases with the square-root of n .**

In the more extreme case that $\alpha = 2$ the maximum degree increases linearly with n .

To put these numbers in perspective, consider a network with one million nodes, and an average degree of $\bar{k} = 3$. If the network follows the exponential degree distribution, the maximum expected degree is only about 10 – not much larger than the average degree.

If the network follows the power-law degree distribution with exponent $\alpha = 2.5$ (recall that this is a typical value for many real-world networks), we get that the maximum degree is about 10,000!



Random vs. Scale-free Networks, Figure 4.6 from networksciencebook.com by Albert-László Barabási

The example above clearly illustrates a major difference between exponential and power-law networks: the latter have nodes with a much greater number of connections than the average node – we typically refer to those nodes as **Hubs**.

The visualization at this page illustrates the difference between exponential and power-law networks focusing on the presence of hubs. The network of major interstate highways in the US follows a Poisson distribution, without any nodes (cities) that have a much larger degree than the average. On the other hand, the network of direct flights between the major US cities follows a power-law degree distribution and there are obvious hubs, such as the airports in Atlanta, Chicago or New York city.

Food for Thought

Derive the integrals shown in this page yourself.